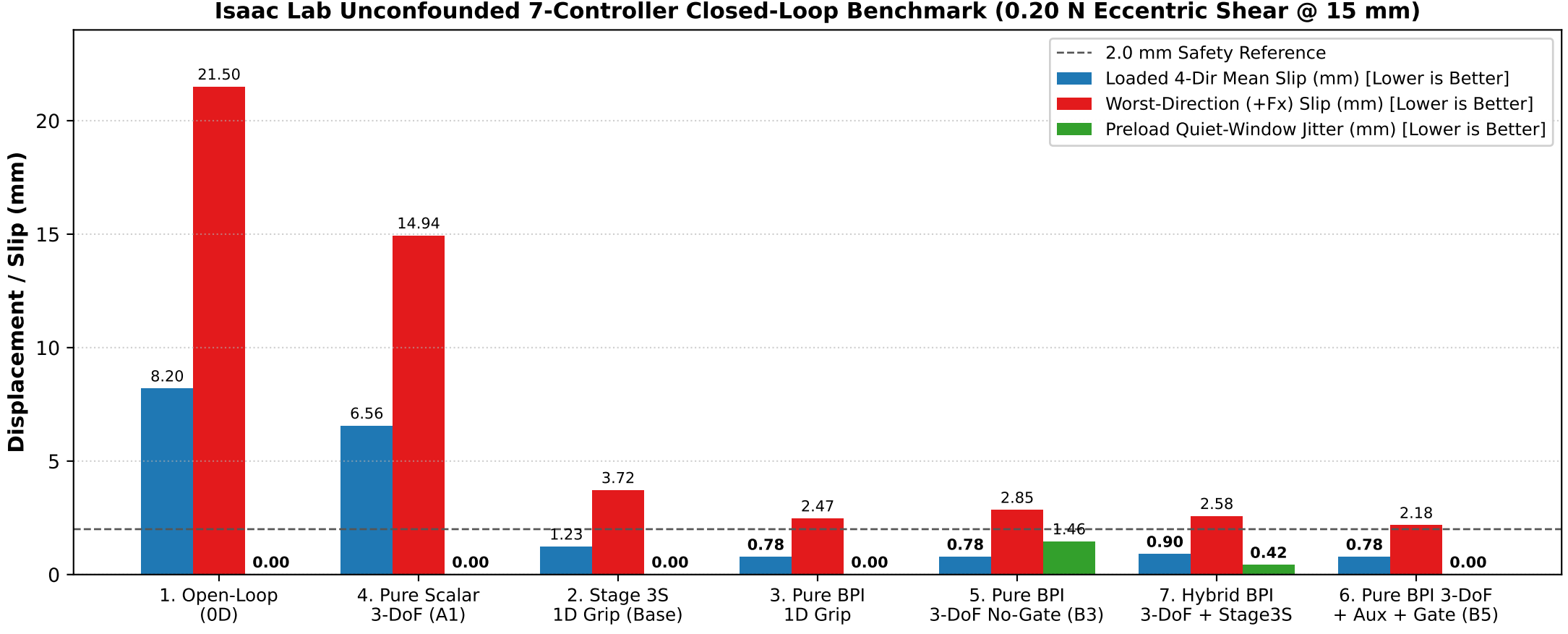
Isaac Lab Unconfounded 7-Controller Closed-Loop Benchmark (0.20 N Eccentric Shear @ 15 mm)
Displacement / Slip (mm)
0
5
10
15
20
2.0 mm Safety Reference
Loaded 4-Dir Mean Slip (mm) [Lower is Better]
Worst-Direction (+Fx) Slip (mm) [Lower is Better]
Preload Quiet-Window Jitter (mm) [Lower is Better]
8.20
21.50
0.00
6.56
14.94
0.00
1.23
3.72
0.00
0.78
2.47
0.00
0.78
2.85
1.46
0.90
2.58
0.42
0.78
2.18
0.00
1. Open-Loop (0D)
4. Pure Scalar 3-DoF (A1)
2. Stage 3S 1D Grip (Base)
3. Pure BPI 1D Grip
5. Pure BPI 3-DoF No-Gate (B3)
7. Hybrid BPI 3-DoF + Stage3S
6. Pure BPI 3-DoF + Aux + Gate (B5)